

EduFin SQL Business Challenge

Phase 4 — Thursday Submission

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STEP 4A — Open Metric Design: "Partner Health Score"

```
WITH inst_summary AS (
    SELECT
        i.institution_id,
        i.institution_name,
        i.institution_type,
        COUNT(*) AS loan_count,
        COUNT(DISTINCT l.customer_id) AS customers,
        SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS default_count
    FROM institutions i
    INNER JOIN loans l
        ON i.institution_id = l.institution_id
    GROUP BY i.institution_id, i.institution_name, i.institution_type
),
multiplier_data AS (
    SELECT
        *,
        (default_count / loan_count) * 100 AS default_rate,
        CASE
            WHEN loan_count >= 10000 THEN 1.5
            WHEN loan_count >= 9500 THEN 1.0
            ELSE 0.5
        END AS volume_multiplier
    FROM inst_summary
)
SELECT
    institution_id AS `Institution Id`,
    institution_name AS `Institution Name`,
    institution_type AS `Institution Type`,
    customers AS `Customers`,
    loan_count AS Loans,
    default_count AS `Default Count`,
    ROUND((1 - default_rate / 100) * volume_multiplier * 100, 2) AS `Health Score`
FROM multiplier_data
ORDER BY `Health Score` DESC;
```

STEP 4B — Default Rate Per Institution

```
WITH default_stats AS (
    SELECT
        i.institution_id,
        i.institution_name,
        COUNT(*) AS loan_count,
        SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS default_count,
        ROUND(default_count * 100.0 / loan_count, 2) AS default_rate
    FROM institutions i
    INNER JOIN loans l
        ON i.institution_id = l.institution_id
    GROUP BY i.institution_id, i.institution_name
    HAVING COUNT(l.loan_id) >= 50
)
SELECT
    institution_id AS `Institution ID`,
```

```

institution_name AS `Institution Name`,
loan_count AS `Loans`,
default_count AS `Default Count`,
CONCAT(default_rate, '%') AS `Default Rate (%)`
FROM default_stats
ORDER BY default_rate DESC, default_count DESC;

```

Institution ID	Institution Name	Loans	Default Count	Default Rate (%)
29	State Commerce Institute	10005	2088	20.87%
33	Central Engineering Institute	9878	2050	20.75%
34	Modern Management Institute	10165	2091	20.57%
31	Regional Commerce University	9915	2039	20.56%
30	National Engineering Institute	9889	2017	20.40%
32	National Commerce Institute	9947	2028	20.39%
50	Government Medical College	9780	1967	20.11%
28	Regional Medical University	10134	1737	17.14%
25	Central Commerce College	10009	1688	16.86%
26	State Arts & Science University	10108	1687	16.69%
24	Modern Law College	10151	1682	16.57%
27	Central Management College	9984	1641	16.44%
23	Advanced Technology University	10121	1652	16.32%
36	Indian Arts & Science University	9897	1324	13.38%
49	Advanced Commerce University	9926	1311	13.21%
8	State Engineering Institute	9860	1300	13.18%
13	State Medical University	10259	1330	12.96%
38	Central Management College	10039	1293	12.88%
12	Central Commerce University	10082	1291	12.80%
21	Central Law University	9956	1273	12.79%
46	Regional Law College	9886	1254	12.68%
7	Indian Law University	10024	1270	12.67%
48	Central Commerce University	9916	1254	12.65%
4	Advanced Arts & Science College	10166	1282	12.61%
2	Central Law University	10107	1274	12.61%

STEP 4C — Financial Loss Attribution

```

WITH financial_exposure AS (
  SELECT
    i.institution_id,
    i.institution_name,
    SUM(l.loan_amount) / 100000000.0 AS loan_amount,
    ROUND(SUM(CASE WHEN l.loan_status = 'Overdue' THEN l.loan_amount * 1.0 ELSE 0
END) / 100000000.0, 2) AS overdue_amount,
    ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0 ELSE
0 END) / 100000000.0, 2) AS default_amount
  FROM institutions i
  INNER JOIN loans l
    ON i.institution_id = l.institution_id
  GROUP BY i.institution_id, i.institution_name
)
SELECT
  institution_id AS `Institution ID`,
  institution_name AS `Institution Name`,
  CONCAT(ROUND(loan_amount, 2), ' Cr') AS `Loan Amount`,
  CONCAT(ROUND(overdue_amount, 2), ' Cr') AS `Overdue Amount`,
  CONCAT(ROUND(default_amount, 2), ' Cr') AS `Default Amount`,
  CONCAT(ROUND(overdue_amount + default_amount, 2), ' Cr') AS `At-Risk Exposure`
FROM financial_exposure
ORDER BY default_amount DESC;

```

	Institution ID	Institution Name	Loan Amount	Overdue Amount	Default Amount	At-Risk Exposure
34		Modern Management Institute	461.35 Cr	34.62 Cr	88.54 Cr	123.16 Cr
31		Regional Commerce University	448.56 Cr	32.84 Cr	85.99 Cr	118.83 Cr
30		National Engineering Institute	451.53 Cr	36.91 Cr	85.15 Cr	122.06 Cr
29		State Commerce Institute	446.11 Cr	37.21 Cr	83.96 Cr	121.17 Cr
33		Central Engineering Institute	447.13 Cr	34.52 Cr	83.5 Cr	118.02 Cr
32		National Commerce Institute	454.88 Cr	36.59 Cr	82.81 Cr	119.4 Cr
50		Government Medical College	437.3 Cr	33.4 Cr	79.94 Cr	113.34 Cr
28		Regional Medical University	454.79 Cr	34.74 Cr	68.86 Cr	103.6 Cr
26		State Arts & Science University	456.33 Cr	38.01 Cr	68.38 Cr	106.39 Cr
25		Central Commerce College	448.52 Cr	37.63 Cr	66.67 Cr	104.3 Cr
24		Modern Law College	450.19 Cr	34.88 Cr	64.85 Cr	99.73 Cr
27		Central Management College	443.59 Cr	36.46 Cr	64.64 Cr	101.1 Cr
23		Advanced Technology University	446.03 Cr	37.48 Cr	63.76 Cr	101.24 Cr
13		State Medical University	459.15 Cr	39.49 Cr	51.87 Cr	91.36 Cr
12		Central Commerce University	457.47 Cr	35.51 Cr	51.42 Cr	86.93 Cr
36		Indian Arts & Science University	448.59 Cr	36.95 Cr	50.59 Cr	87.54 Cr
38		Central Management College	457.73 Cr	38.54 Cr	50.51 Cr	89.05 Cr
7		Indian Law University	451.27 Cr	35.6 Cr	49.63 Cr	85.23 Cr
4		Advanced Arts & Science College	451.9 Cr	34.66 Cr	49.19 Cr	83.85 Cr
21		Central Law University	453.34 Cr	33.36 Cr	49.09 Cr	82.45 Cr
2		Central Law University	452.22 Cr	36.94 Cr	48.92 Cr	85.86 Cr
49		Advanced Commerce University	445.75 Cr	36.62 Cr	48.64 Cr	85.26 Cr
14		Indian Technology University	455.77 Cr	35.94 Cr	48.56 Cr	84.5 Cr
48		Central Commerce University	442.78 Cr	35.18 Cr	48.51 Cr	83.69 Cr
5		Regional Law University	440.85 Cr	33.14 Cr	48.43 Cr	81.57 Cr

STEP 4D — Minimum Data Threshold Gate

```

WITH default_stats AS (
  SELECT
    i.institution_id,
    i.institution_name,
    i.institution_type,
    COUNT(*) AS loan_count,
    SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS default_count,
    ROUND(default_count * 100.0 / loan_count, 2) AS default_rate
  FROM institutions i
  INNER JOIN loans l
    ON i.institution_id = l.institution_id
  GROUP BY i.institution_id, i.institution_name, i.institution_type
  HAVING COUNT(l.loan_id) >= 50
)
SELECT
  institution_id AS `Institution ID`,
  institution_name AS `Institution Name`,
  institution_type AS `Institution Type`,
  CONCAT(default_rate, '%') AS `Default Rate (%)`,
  CASE
    WHEN default_rate > 20 THEN 'Critical Risk'
    WHEN default_rate > 15 THEN 'High Risk'
    WHEN default_rate > 10 THEN 'Monitor'
    ELSE 'Safe'
  END AS `Risk Classification`
FROM default_stats
ORDER BY default_rate DESC;

```

	Institution ID	Institution Name	Institution Type	Default Rate (%)	Risk Classification
	29	State Commerce Institute	Institute	20.87%	Critical Risk
	33	Central Engineering Institute	Institute	20.75%	Critical Risk
	34	Modern Management Institute	Institute	20.57%	Critical Risk
	31	Regional Commerce University	University	20.56%	Critical Risk
	30	National Engineering Institute	Institute	20.40%	Critical Risk
	32	National Commerce Institute	Institute	20.39%	Critical Risk
	50	Government Medical College	College	20.11%	Critical Risk
	28	Regional Medical University	University	17.14%	High Risk
	25	Central Commerce College	College	16.86%	High Risk
	26	State Arts & Science University	University	16.69%	High Risk
	24	Modern Law College	College	16.57%	High Risk
	27	Central Management College	College	16.44%	High Risk
	23	Advanced Technology University	University	16.32%	High Risk
	36	Indian Arts & Science University	University	13.38%	Monitor
	49	Advanced Commerce University	University	13.21%	Monitor
	8	State Engineering Institute	Institute	13.18%	Monitor
	13	State Medical University	University	12.96%	Monitor
	38	Central Management College	College	12.88%	Monitor
	12	Central Commerce University	University	12.80%	Monitor
	21	Central Law University	University	12.79%	Monitor
	46	Regional Law College	College	12.68%	Monitor
	7	Indian Law University	University	12.67%	Monitor
	48	Central Commerce University	University	12.65%	Monitor
	4	Advanced Arts & Science College	College	12.61%	Monitor
	2	Central Law University	University	12.61%	Monitor

STEP 4E — Partnership Review Dashboard (Final Output)

```

WITH default_stats AS (
  SELECT
    i.institution_id,
    i.institution_name,
    i.institution_type,
    COUNT(*) AS loan_count,
    SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS default_count,
    ROUND(default_count * 100.0 / loan_count, 2) AS default_rate
  FROM institutions i
  INNER JOIN loans l
    ON i.institution_id = l.institution_id
  GROUP BY i.institution_id, i.institution_name, i.institution_type
)
SELECT
  institution_id AS `Institution ID`,
  institution_name AS `Institution Name`,
  institution_type AS `Institution Type`,
  CONCAT(default_rate, '%') AS `Default Rate (%)`,
  CASE
    WHEN default_rate > 15 THEN 'Blacklist'
    WHEN default_rate > 10 THEN 'Monitor'
    ELSE 'Expand'
  END AS `Partner Flag`
FROM default_stats
ORDER BY default_rate DESC;

```

Institution ID	Institution Name	Institution Type	Default Rate (%)	Partner Flag
29	State Commerce Institute	Institute	20.87%	Blacklist
33	Central Engineering Institute	Institute	20.75%	Blacklist
34	Modern Management Institute	Institute	20.57%	Blacklist
31	Regional Commerce University	University	20.56%	Blacklist
30	National Engineering Institute	Institute	20.40%	Blacklist
32	National Commerce Institute	Institute	20.39%	Blacklist
50	Government Medical College	College	20.11%	Blacklist
28	Regional Medical University	University	17.14%	Blacklist
25	Central Commerce College	College	16.86%	Blacklist
26	State Arts & Science University	University	16.69%	Blacklist
24	Modern Law College	College	16.57%	Blacklist
27	Central Management College	College	16.44%	Blacklist
23	Advanced Technology University	University	16.32%	Blacklist
36	Indian Arts & Science University	University	13.38%	Monitor
49	Advanced Commerce University	University	13.21%	Monitor
8	State Engineering Institute	Institute	13.18%	Monitor
13	State Medical University	University	12.96%	Monitor
38	Central Management College	College	12.88%	Monitor
12	Central Commerce University	University	12.80%	Monitor
21	Central Law University	University	12.79%	Monitor
46	Regional Law College	College	12.68%	Monitor
7	Indian Law University	University	12.67%	Monitor
48	Central Commerce University	University	12.65%	Monitor
4	Advanced Arts & Science College	College	12.61%	Monitor
2	Central Law University	University	12.61%	Monitor

Custom Column

```

CASE
  WHEN default_rate > 15 THEN 'Blacklist'
  WHEN default_rate > 10 THEN 'Monitor'
  ELSE 'Expand'
END AS `Partner Flag`

CASE WHEN EXISTS (
  SELECT 1 FROM customers c
  JOIN loans l
    ON c.customer_id = l.customer_id
  WHERE l.institution_id = institution_summary.institution_id AND c.cibil_score >
900
)

```

A fixed threshold like `default_rate > 15%` assumes you already know what "bad" looks like before seeing the data, but that number is arbitrary and breaks down across different economic conditions, loan types, or time periods. Comparing against the portfolio average is self-calibrating: if the whole portfolio is performing poorly (say average is 18%), a 15% institution is actually your best partner, not a suspension candidate. It also means your flags automatically adapt as the portfolio changes — no one needs to manually update hardcoded thresholds every quarter.

WHY Gate

1. The institution with 0 defaults — what action did you assign it, and why? Write your classification AND your reasoning. ("I assigned it X because Y, not Z because A.")

'National Law Institute' (Institution ID 17) is the only institution with 0 defaults. I classified the institution as 'Reward' and not 'Retain', 'Review' OR 'Suspend', because its default rate of 0.00% falls below 0.5x the portfolio average, making it the strongest performer/institution in the entire dataset. I did not classify it as 'Retain' because Retain is reserved for institutions between 0.5x and 1x the average, which still carry moderate risk. 'Review' and 'Suspend' are ruled out entirely since those require a default rate at or above the portfolio average.

2. Write your Suspension Justification for the top-ranked institution to be suspended. Format it as if going to a legal review: Institution name, default rate, loan count, total defaulted exposure, which threshold was exceeded, recommended action, and what monitoring must happen post-suspension.

- Institution: State Commerce Institute (Institution ID: 29) - Default Rate: 20.87% - Total Loans Disbursed: 10005 - Total Defaulted Exposure: 83.96 Crores - Threshold Exceeded: 20.87% exceeds the 1.5x portfolio average threshold - Recommended Action: Immediate suspension of new loan disbursements pending full portfolio audit. No new student loan applications to be processed through this institution until review is complete. - Post-Suspension Monitoring Requirements: - 83.96 Cr in defaulted exposure must be assigned to collections with a 90-day recovery report due to the Partnership Director - Institution returned Data Clean on the anomaly check, meaning no CIBIL >900 irregularities were detected; however a manual data audit is still recommended given the volume - Partnership agreement to be placed under legal hold; no contract renewals or amendments permitted during suspension period.

3. If you were the Partnership Director and received your final output — what is the FIRST call you make? Who do you call, about which institution, and what specifically do you say in that call?

If I were the partnership director that receive my own final output, then I would first try to call the institution who had 0% default rate despite having 10026 loans. This result did not come from a rounding error, but due to the structural difference of how you guys are operating compared to the other institutions. Before I spend the next few weeks on suspension hearings for the bottom of this portfolio, I want to understand what you're doing differently like, what's your screening process, your student selection criteria, your collections touchpoints, whatever it is. Because if we can document it and replicate it, that's worth more to this organization than suspending 10 bad partners.